Target-Edge Leakage Inflates Topological Link Prediction: A Quantification and a Corrected Protocol

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Abstract

Persistent homology is increasingly used to build node and neighborhood features for graph learning. When those features are computed once on the full graph and then reused to predict individual edges, the feature of a node already encodes the very edge being predicted, and the reported accuracy is inflated. I quantify this target-edge leakage for topological link prediction across ten graphs, six real, including citation, coauthorship, and literary networks, and four synthetic, spanning scale-free, community, small-world, and spatial topologies. Under a protocol that removes each fold's test edges from the graph before any persistence feature is computed, apparent performance falls sharply and systematically: the mean inflation of the naive protocol is 0.11 AUC, reaches 0.20 on the citation graph, and on several graphs drops corrected single-parameter performance to chance, with a 95 percent confidence interval on the inflation that excludes zero. The inflation is present on all ten graphs and for both single-parameter and multi-parameter features, and on a majority of graphs it reverses which of two methods appears better. As a controlled application of the protocol I compare single-parameter and multi-parameter fibered persistence, and find that once leakage is removed the multi-parameter advantage for edge prediction is small but statistically significant across the ten graphs, on the order of 0.01 to 0.03 AUC and concentrated where single-parameter persistence is weak. The reported performance of topological link prediction depends more on whether the protocol controls target-edge leakage than on the choice of persistence invariant, and I recommend the corrected protocol as a default for this literature.

Keywords: topological data analysis, persistent homology, link prediction, data leakage, evaluation protocol, multi-parameter persistence, fibered barcodes

1. Introduction

Topological data analysis, and persistent homology in particular, has become a productive source of features for graph learning. Persistence computed on a graph or on the neighborhoods of its nodes yields multiscale descriptors that feed classifiers for node classification, graph classification, and

link prediction. In classification the correct procedure is to compute persistence on the whole graph, because no individual target is withheld from the topology.

Link prediction is different, and the difference is easy to miss. When one predicts whether an edge exists between two nodes, and the node features were computed on the full graph, the feature of each endpoint already reflects the presence of the candidate edge: for a true edge the other endpoint sits in the neighborhood used to build the feature, while for a non-edge it does not. A classifier can then separate positives from negatives on a property that is a restatement of the label rather than a prediction of it. This is target-edge leakage. It is understood in the graph-learning community, where the remedy is to hide test edges before featurization, but the habit of computing persistence once on the full graph, carried over from classification where it is correct, makes topological link-prediction pipelines prone to it.

This paper quantifies the severity of that leakage for persistence-based link prediction across ten graphs, weighted toward real networks and spanning distinct generative families, and shows that it is large, systematic, and able to alter not only the magnitude of reported performance but the relative ranking of methods. It then uses the corrected protocol to answer a question the multi-parameter persistence literature usually assumes rather than tests, whether multi-parameter persistence actually beats single-parameter for edge prediction, and reports what that comparison yields once leakage is removed. The contributions are a quantification of leakage severity in the persistence setting, a corrected protocol recommended as a default, and a leakage-controlled comparison of single-parameter and multi-parameter persistence.

2. Related Work

Leakage from improper edge splitting is documented in the graph-representation-learning literature, and hiding test edges before computing structural features is the accepted discipline for link prediction. The contribution here is not to identify leakage in general but to quantify its severity for persistence-based features, where the whole-graph featurization habit is often imported from graph classification, and to show its consequences for method comparison. The persistence tools used are standard: stable vector summaries of diagrams (Adams et al. 2017), fibered barcodes that restrict a multi-parameter module to lines with single-parameter stability along each slice (Lesnick and Wright 2015), and the established multi-parameter descriptors for learning (Carriere and Blumberg 2020; Loiseaux et al. 2023). Node filtration functions draw on Ollivier-Ricci curvature (Ollivier 2009; Jost and Liu 2014) and the path Laplacian (Wang and Wei 2023).

3. Target-Edge Leakage and the Corrected Protocol

Mechanism. Let each node carry a feature computed from its neighborhood in a graph G. For a candidate edge between nodes u and v, leakage arises because, when G contains the edge, v lies in the neighborhood of u and contributes to u's feature, and vice versa. Positive candidates therefore differ from negative candidates in a way that is a direct consequence of the label, independent of any genuine structural signal. Because this holds for any neighborhood-based feature, it affects

single-parameter and multi-parameter persistence alike, so it does not by itself distort their comparison, but it inflates every absolute number and can shift comparisons through second-order effects.

Corrected protocol. I adopt the standard graph-autoencoder discipline.[1] For each cross-validation fold, I remove the fold's test positive edges from G, recompute all node features on the residual graph, and only then train on the training edges and evaluate on the held-out test edges. Test positive edges are absent from the graph used to featurize them; non-edges are absent in any case. The trivial edge-in-neighborhood signal is therefore unavailable at test time. The naive protocol I compare against is identical except that features are computed once on the full graph and reused across folds.

Setup. Each node carries three standardized filtration functions computed on its bounded local neighborhood: a connectivity or strength term, mean Ollivier-Ricci curvature, and the path-Laplacian spectral gap of the local subgraph. Single-parameter persistence uses one filtration; the multi-parameter representation is a fibered barcode obtained by slicing a scale-by-function bifiltration along a family of lines. I evaluate with two capacity-controlled classifiers, L2-regularized logistic regression and gradient-boosted trees, under three-fold stratified cross-validation with balanced negative sampling. Label-shuffled runs return a mean AUC near 0.49, confirming that observed effects are not memorization. The ten graphs comprise six real networks, the IEEE 118-bus power system, the Les Miserables co-occurrence network, a citation graph (Citeseer), a coauthorship graph (Netscience), and two social networks (a dolphin association network and Zachary's karate club), together with four synthetic families: Watts-Strogatz, Barabasi-Albert, stochastic-block, and random-geometric.

4. How Much Leakage Inflates Results

Table 1 reports gradient-boosted AUC under the naive and corrected protocols, for single-parameter persistence and for the multi-parameter fibered representation, on all ten graphs.

Table 1. AUC under the naive (leaky) and corrected (rigorous) protocols, and the inflation (naive minus corrected).

Graph	Domain	Single-parameter persistence			Multi-parameter fibered		
		Leaky	Corrected	Inflation	Leaky	Corrected	Inflation
Citeseer	citation	0.788	0.590	+0.198	0.810	0.633	+0.178
Netscience	coauthorship	0.892	0.748	+0.144	0.886	0.753	+0.133
Les Miserables	literary	0.887	0.827	+0.060	0.886	0.830	+0.055
IEEE 118	power grid	0.769	0.602	+0.167	0.819	0.594	+0.225
Dolphins	animal social	0.776	0.672	+0.104	0.740	0.695	+0.045
Karate	social club	0.854	0.812	+0.042	0.825	0.801	+0.024

[1]Hiding test edges from the graph before featurization is established practice in graph representation learning, where link-prediction models are routinely trained and evaluated on an observed graph from which the held-out edges have been removed. The point of this paper is not to identify leakage as a novel phenomenon but to quantify its consequences for persistence-based features specifically, where the habit of computing persistence once on the full graph is commonly carried over from graph classification, in which it is correct.

Graph	Domain	Single-parameter persistence			Multi-parameter fibered		
		Leaky	Corrected	Inflation	Leaky	Corrected	Inflation
Barabasi-Albert	scale-free	0.646	0.471	+0.175	0.627	0.486	+0.141
Stochastic block	community	0.662	0.562	+0.099	0.676	0.603	+0.073
Watts-Strogatz	small-world	0.610	0.537	+0.073	0.638	0.562	+0.076
Random geometric	spatial	0.869	0.757	+0.113	0.847	0.761	+0.087

The inflation is present on every graph and for both representations. It averages 0.12 AUC for single-parameter persistence and 0.10 for the multi-parameter representation, with a 95 percent confidence interval on the single-parameter inflation of [+0.085, +0.150] that excludes zero, and it reaches 0.20 on the citation graph. On several graphs, including the scale-free, small-world, and community graphs, the corrected single-parameter AUC falls to chance or below, and on the citation graph single-parameter persistence drops from an apparent 0.79 to 0.59. Even on graphs where the corrected performance is genuinely strong, such as Les Miserables, roughly a twentieth of an AUC point is leakage. A practitioner reporting the naive numbers would substantially overstate what the topology predicts.

Leakage also alters relative conclusions, though more mildly than it alters absolute ones. Because the naive protocol inflates single-parameter persistence slightly more than the multi-parameter representation on average, it compresses the apparent gap between them; the corrected protocol widens or restores that gap on most graphs, and on six of the ten it reverses the sign of which representation appears better, generally by inflating single-parameter persistence enough under the naive protocol to hide the small multi-parameter advantage. The dominant effect of leakage is inflation of absolute performance, but its capacity to change the ranking of methods is not hypothetical.

5. A Leakage-Controlled Comparison of Single- and Multi-Parameter Persistence

With a protocol that controls leakage, the downstream question can be asked honestly. Table 2 reports the per-graph gain of the multi-parameter fibered representation over single-parameter persistence under the corrected protocol, for both classifiers.

Table 2. Corrected-protocol gain of multi-parameter over single-parameter persistence (AUC).

Graph	Domain	Logit gain	GBM gain
Citeseer	citation	+0.051	+0.042
Netscience	coauthorship	+0.009	+0.004
Les Miserables	literary	+0.038	+0.003
IEEE 118	power grid	+0.064	-0.009
Dolphins	animal social	+0.012	+0.023
Karate	social club	+0.049	-0.011
Barabasi-Albert	scale-free	-0.008	+0.015
Stochastic block	community	+0.032	+0.041

Graph	Domain	Logit gain	GBM gain
Watts-Strogatz	small-world	+0.069	+0.025
Random geometric	spatial	-0.022	+0.004

Treating each graph as one replication, the mean gain is +0.030 AUC on the linear probe (95 percent confidence interval [+0.011, +0.048]) and +0.014 on gradient boosting (interval [+0.002, +0.025]), positive in eight of ten domains on both probes. One-sample t-tests against zero give p of 0.007 and 0.023, and Wilcoxon signed-rank tests give 0.010 and 0.032. By these tests the multi-parameter advantage is statistically significant, but it is small, on the order of one to three hundredths of an AUC point, and its significance owes as much to the size of the panel as to the size of the effect: the same small gain was not distinguishable from zero on a six-graph subset. The consistent pattern is that the increment is largest where single-parameter persistence is weak, for instance the citation and community graphs, and negligible or slightly negative where single-parameter persistence already predicts links, such as the karate club and the power grid on the boosted probe, which suggests that multi-parameter structure supplies information mainly when the dominant single axis is uninformative. This advantage is not an artifact of removing leakage; leakage was masking rather than manufacturing it, since the corrected gap exceeds the naive one on most graphs. A lightweight diagram-statistics vectorization of the fibered barcodes was competitive with the multi-parameter persistence image throughout, so the small advantage that does appear does not depend on the heavier vectorization.

6. Discussion

The primary finding is a caution the topological-link-prediction literature should absorb. Computing persistence features on the full graph, correct for classification, leaks the target edge in link prediction and inflates reported AUC by a mean of more than a tenth of a point across these ten graphs, by about a fifth of a point in the worst case, and enough to turn a chance-level result into an apparently strong one. Any claim that persistence predicts links should be made under a protocol that hides the target edge, and I recommend the corrected protocol of Section 3 as a default. The effect is large enough that comparisons of published topological link-prediction results are unreliable unless their protocols are known to control it.

The controlled comparison delivers a more measured secondary message. The often-assumed advantage of multi-parameter over single-parameter persistence, once measured under a leakage-free protocol and across a diverse panel of graphs, is real and statistically significant but small, and it appears mainly where the single dominant filtration axis is uninformative. This is a weaker endorsement than the multi-parameter literature usually implies: the gain is a fraction of the leakage it is dwarfed by, it is inconsistent enough to vanish or reverse on individual graphs, and it emerges as significant only with enough replications to resolve a small effect. It argues not against multi-parameter persistence but against assuming its benefit for a given task without measuring it, and for reporting effect sizes and cross-domain variability rather than a single favorable benchmark.

7. Limitations

The graph panel, while spanning ten networks and weighted toward real data, is bounded in size and density: to keep the per-fold recomputation of curvature tractable I capped node neighborhoods, used three-fold rather than five-fold cross-validation, and excluded the densest real graphs, whose corrected-protocol cost was prohibitive. The panel therefore skews toward sparser graphs, and a larger, denser, and more heavily real-world panel would sharpen both the leakage confidence interval and the ten-replication significance test, which, though now adequately powered to detect the small multi-parameter effect, remains modest. The evaluation is inductive in edges but transductive in nodes, so nodes recur across folds even though target edges do not. The filtration functions and the fibered-barcode construction are particular choices among many, and other choices might change the cross-domain behavior of the comparison in Section 5. The leakage quantification of Section 4, by contrast, follows from the mechanism and is unlikely to be specific to these choices.

8. Conclusion

Target-edge leakage inflates topological link prediction substantially and systematically, by a mean of about 0.11 AUC and by roughly 0.20 in the worst case here, and it can reverse which of two methods appears better. The corrected protocol that hides test edges before featurization removes the inflation and should be standard. Under that protocol, the advantage of multi-parameter fibered persistence over single-parameter persistence for edge prediction is statistically significant but small and domain-dependent, concentrated where single-parameter persistence is weak. The reported performance of topological link prediction depends more on protocol than on invariant, and I report these results, including the modest and unflattering ones, so that the field can calibrate accordingly.

Declaration of Competing Interests

The author holds U.S. provisional patent applications related to topological feature computation on directed graphs. These interests do not affect the analysis, which is conducted on publicly available and openly generated data using standard, reproducible methods, and which reports negative and null results.

Data and Code Availability

All graphs are publicly available or generated from documented random seeds. The evaluation harness implementing both the naive and corrected protocols, and the analysis reproducing every reported number, are available from the author.

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